55 Creating an International FLOSS Web Platform to Share Archaeological Data

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The current paper presents the design and architecture of an international FLOSS web platform for communicating archaeological data in 3D. According to the findings of the state-of-the-art analysis obtained from the evaluation of around 50 case studies of web-based 3D archaeological visualizations introduced at Archeo.FOSS 19, the goal of the project is to develop a prototype for the exploration, analysis, and distribution of archaeological heritage in accordance with the Open Science paradigm and the FAIR principles.

This infrastructure allows combining real-time 3D web visualization (excavation stratigraphy, architectural meshes), 2D spatial layers (raster/vector GIS data), and field documentation. The platform, designed for the specialized research community and for the public use, provides a web-based query interface and direct downloading of the raw data.

In this paper, we will discuss the selection and benchmarking of FLOSS web frameworks, system architecture, and initial deployment results based on the Archeologia a Cervia dataset. Moreover, the platform is open to other datasets provided by volunteers for testing and development of an efficient data management pipeline in the application.